Modeling Crash Injury Severity in New York City: A Linear Regression and Simulation Study

ST 453 Midterm Project | March 10, 2026

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1. Data Description

The data set analyzed in this project is the New York City Motor Vehicle Collisions – Crashes data set, a publicly available collection of police-reported traffic crashes maintained by the NYC Department of Transportation. This is public data freely available from NYC OpenData at:

https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95/about_data

The full dataset contains 2,247,203 collisions (obs) and 29 variables. Variables include crash time, borough, injury and fatality counts by road-user type (pedestrians, cyclists, motorists), up to five contributing factors, and up to five vehicle type codes. The primary response variable is NUMBER.OF.PERSONS.INJURED, a non-negative integer recording total persons injured per crash. This data is updated daily, automatically.

To construct an analysis model, six variables were chosen and retained: CRASH.TIME, BOROUGH, NUMBER.OF.PERSONS.INJURED, CONTRIBUTING.FACTOR.VEHICLE.1, VEHICLE.TYPE.CODE.1, and VEHICLE.TYPE.CODE.2. Rows with missing or blank values in any used field were removed, and a random sample of 20,000 crashes was drawn using set.seed(1). After constructing the regression design matrix and removing incomplete rows, the final analysis sample contains 19,999 observations and 15 model columns (including the intercept). Table 1 summarizes data origin. This data set is well suited for studying crash severity because it records both an injury outcome and rich crash-level contextual information whose practical

relevance extends to transportation safety policy and prevention.

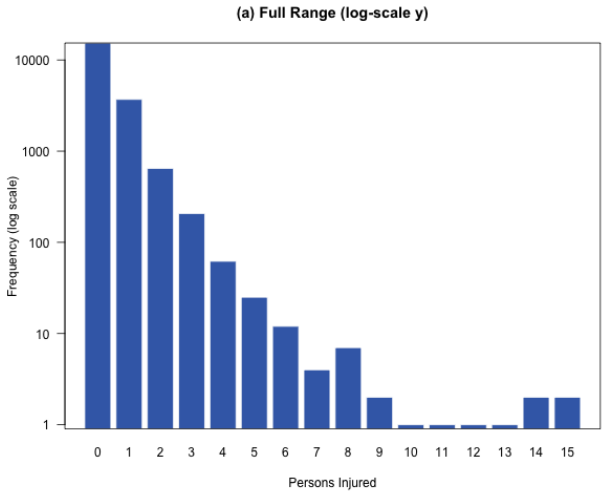
Statistic	Value
Minimum	0
1st Quartile	0
Median	0
Mean	0.3069
3rd Quartile	0
Maximum	15
Std. Deviation	0.6719
% Crashes with 0 Injuries	~78.6%

Predictor	Count	Proportion	Pred	Count	Prop
Night (9PM–5AM)	3,860	19.3%	**Following Too Closely**	730	3.7%
Multi-Vehicle	15,873	79.4%	**Failure to Yield**	1,233	6.2%
Brooklyn	6,474	32.4%	**Unspecified Factor**	7,422	37.1%
Manhattan	4,340	21.7%	**Pickup**	417	2.1%
Queens	5,398	27.0%	**Motorcycle**	176	0.9%
Bronx	2,964	14.8%	**Sedan**	5,865	29.3%
Staten Island (reference)	923	4.6%	**SUV**	6,115	30.6%
Inattention/Distraction	**3,932**	**19.7%**			

Table 1. Summary Statistics for NUMBER.OF.PERSONS.INJURED (n = 19,999)

2. Summary Statistics and Exploratory Analysis

In the cleaned sample, NUMBER.OF.PERSONS.INJURED has minimum 0, first quartile 0, median 0, mean 0.3069, third quartile 0, and maximum 15. The distribution is strongly right skewed with many collisions at zero. Most crashes involve no injuries. This zero-heavy feature is an important characteristic of the data, it will make the outcome difficult to model with a simple linear mean structure. Various predictor categories are imbalanced: 3,860 nighttime crashes, 15,873 multi-vehicle crashes, and only 176 motorcycle crashes, rarer categories yield more variable estimates. Figures 1–4 show the distribution of the response and exploratory comparisons across crash characteristics, motivating the predictors included in the regression model.



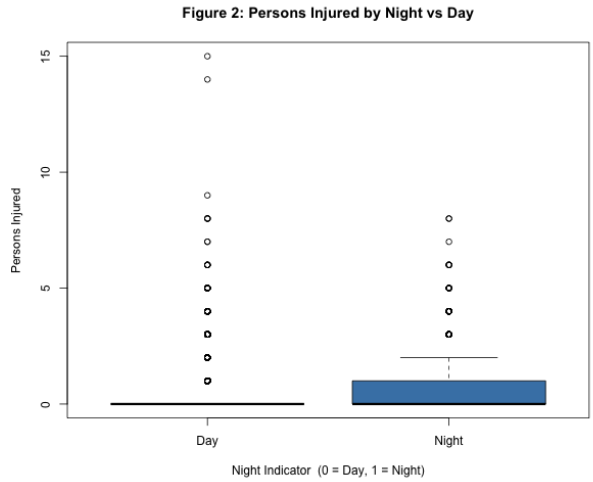
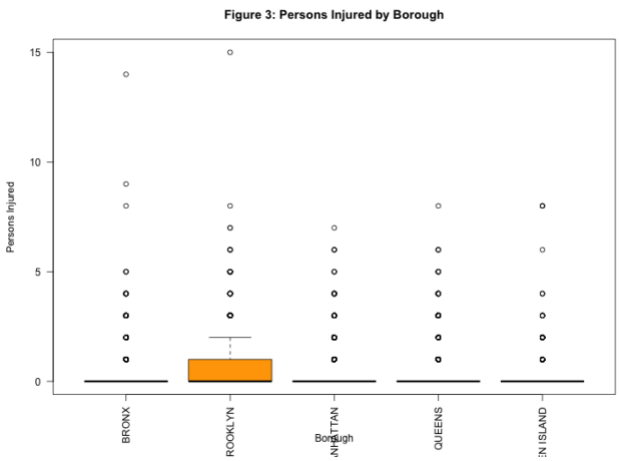
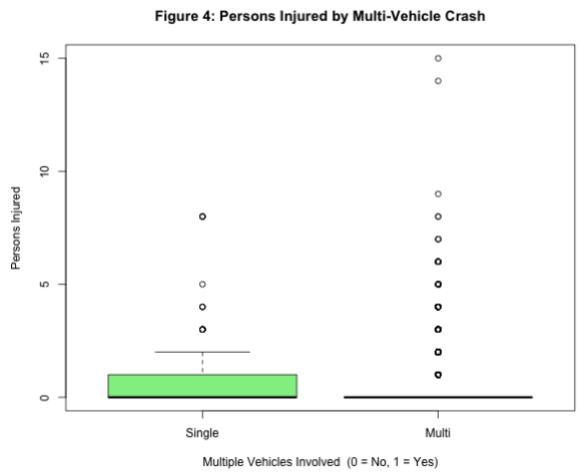


Figure 1. Distribution of NUMBER.OF.PERSONS.INJURED (n=19,999). Strongly right-skewed with large mass at zero





The boxplots reveal meaningful patterns. Nighttime crashes trend toward slightly higher injury counts. Brooklyn has the highest average injury count among boroughs. Multi-vehicle involvement alters the injury distribution compared to single-vehicle crashes. These patterns collectively motivate the variables included in the regression.

3. Variables of Interest and Relationships Being Studied

The primary outcome is the number of persons injured per crash, used as a continuous measure of crash severity. The inferential goal is to estimate how the conditional mean varies with observable crash characteristics. The Night indicator (1 = crash between 9 PM and 5 AM) reflects reduced visibility and driver fatigue. MultiVehicle (1 = second vehicle type recorded) captures multi-vehicle involvement. Borough indicators for Brooklyn, Manhattan, Queens, and Bronx (reference: Staten Island) account for geographic differences in traffic density and infrastructure.

Contributing factor indicators cover Inattention (Driver Inattention/Distraction), FollowingTooClosely, FailureToYield (Failure to Yield Right-of-Way), and UnspecifiedFactor. Vehicle-type indicators cover Sedan, SUV, Pickup, and Motorcycle. The central questions are: (1) Are nighttime crashes associated with higher expected injury counts? (2) Which contributing

factors and vehicle types of link to greater severity? (3) Do these relationships hold after controlling for other crash characteristics?

4. Statistical Model

Let Y_i be the number of persons injured in crash *i* and let X_i be the 15-dimensional covariate vector (including intercept). The model is:

$$Y_i = \beta_0 + \beta_1 Night_i + \beta_2 MultiVehicle_i + \beta_3 Brooklyn_i + \cdots + \beta_{14} Motorcycle_i + U_i$$

or in matrix form $Y = X\beta + U$, where errors U_i are independent with mean zero and variance σ^2. The model is a deliberate simplification: the outcome is a zero-inflated count, so the linear specification does not perfectly capture the probability structure of Y. Its value lies in providing an interpretable approximation to the conditional mean $E(Y_i|X_i) = X_i^T\beta$.The parameters of interest are $\beta \in \mathbb{R}^{15}$ and $\sigma^2 > 0$.

5. Generating Synthetic Data from the Model

To generate a synthetic data set, given true values β^{true}, σ^{true}, and fixed design matrix X: (1) draw $\epsilon_i \sim N(0, (\sigma^{true})^2)$ for i = 1, …, n; (2) form $Y^{(sim)} = X\beta^{true} + \epsilon$. The real-data design matrix X is held fixed so synthetic data preserve the same predictor structure, sample size (n=19,999), and covariate distribution as the real analysis. True parameter values are set to the real-data estimates: $\beta^{true} = \hat{\beta}$ *and* $\sigma^{true} = \hat{\sigma} = 0.6718892$. This standard approach calibrates the simulation to the actual data, making results directly relevant to real-data estimation.

6. Computational Approach to Fitting the Model

Coefficients are estimated by ordinary least squares (OLS): $\hat{\beta} = \arg\min_{\beta \in \mathbb{R}^p} ||Y - X\beta||_2^2$. In practice this uses QR decomposition via qr.solve(X,y) in R, numerically stable, no matrix inversion. Fitted

values $\hat{Y} = X\hat{\beta}$, residuals $\hat{U} = Y - \hat{Y}$, and residual standard deviation $\hat{\sigma} = 0.6719$ are used as the noise scale in the simulation. Standard errors use the diagonal of $(X^T X)^{-1}\hat{\sigma}^2_{unb}$where computed via solve(t(X)%*%X).

7. Simulation Study

A simulation study was conducted to assess whether OLS accurately recovers true population parameters from synthetic data. Each of N=1,000 replications: (1) generates $Y^{(sim)} = X\hat{\beta} + \epsilon \; with \; \epsilon \sim N(0, \hat{\sigma}^2 I_n)$; (2) refits OLS via qr.solve; (3) records estimated coefficients and in-sample RMSE; then (4) sampling distributions are summarized with plots and tables.

While remaining practical, this provides smooth sampling distributions and stable Monte Carlo summaries of bias and standard deviation. Fewer replications would yield noisy histograms and unreliable bias estimates. Each synthetic data set uses the same design matrix X (n=19,999, p=15). Because estimator performance depends on both sample size and model complexity, matching the real-data setting exactly makes simulation results directly relevant.

OLS performed very well under the assumed model. Mean estimated coefficients across 1,000 replications were very close to the true generating values. Night: true=0.08195, mean estimate = 0.08170 (bias = −0.00025). MultiVehicle: true = −0.10504, mean = −0.10463 (bias = 0.00040). FailureToYield: true = 0.25762, mean = 0.25812 (bias = +0.00050). Motorcycle: true = 0.32562, mean = 0.32579 (bias = +0.00017). All biases are negligible relative to Monte Carlo standard deviations, confirming the estimator is approximately unbiased. Simulation RMSE: mean = 0.6717, SD = 0.0033, closely matching $\hat{\sigma} = 0.6719$. Table 4 presents the full summary.

Parameter	True_Value	Mean_Estimate	Bias	SD_Estimate
Intercept	0.36679306	0.36673431	-5.875E-05	0.02894491
Night	0.08195054	0.08126258	-0.00068796	0.01207199
MultiVehicle	-0.10503618	-0.10428531	0.00075087	0.01226535
Brooklyn	0.03272208	0.03206106	-0.00066101	0.02564958
Manhattan	-0.08310143	-0.08380701	-0.00070559	0.02622135
Queens	-0.02660447	-0.02804772	-0.00144325	0.02560659
Bronx	0.00418753	0.00365921	-0.00052833	0.02655071
Inattention	0.05497	0.05501653	4.654E-05	0.01411421
FollowingTooClosely	0.07774905	0.07819864	0.00044959	0.02566785
FailureToYield	0.25762122	0.25850944	0.00088822	0.02103538
UnspecifiedFactor	-0.0347808	-0.0344336	0.0003472	0.01200059
Sedan	0.01165291	0.01231501	0.0006621	0.01207237
SUV	8.89E-05	0.00037089	0.00028199	0.01217093
Pickup	-0.06035272	-0.06184222	-0.0014895	0.03391359
Motorcycle	0.32561859	0.32580906	0.00019047	0.05151316

Table 4. Simulation summary over N=1,000 replications for all 15 parameters.

The Monte Carlo mean is within a fraction of a standard deviation of the generating value for all coefficients. Figures 6–7 show sampling distributions for Night and MultiVehicle both of which are tightly centered on the reference lines. Figure 9 places all 15 points on or very near the 45-degree line. Nothing went wrong, the least squares algorithm is correctly recovering the true parameter values.

8. Hypothesis Tests and Confidence Intervals

Inference was conducted for the Night coefficient. Let β_{Night} be the population coefficient. Test: $H_0: \beta_{Night} = 0 \; vs. H_A: \beta_{Night} \neq 0$. The estimated coefficient is $\hat{\beta}_{Night} = 0.08195054$, with $SE = 0.01207275$. The t-statistic is $0.08195054/0.01207275 \approx 6.79$ on 19,984 df. With an extremely small p-value the null is rejected; there's strong evidence that nighttime crashes relate to a nonzero change in expected injury count. The 95% confidence interval for β_{Night} is $(0.05828696, 0.10561413)$. Since this interval is entirely positive and does not contain zero, nighttime crashes are associated with higher expected injury counts, with other predictors fixed. Table 3 presents the full inference table for all 15 parameters. Inattention, FollowingTooClosely, FailureToYield, and Motorcycle are significant with positive signs. MultiVehicle, Manhattan, and

UnspecifiedFactor are significant with negative signs. Brooklyn, Queens, Bronx, Sedan, SUV, Pickup are not statistically noticeable from zero at the 0.05 level in this specification.

Parameter	Estimate	Std_Error	t_stat	p_value	CI_Lower	CI_Upper
Intercept	0.366793	0.027793	13.197164	0	0.312316	0.42127
Night	0.081951	0.012073	6.788061	0	0.058287	0.105614
MultiVehicle	-0.105036	0.012061	-8.708673	0	-0.128677	-0.081395
Brooklyn	0.032722	0.024883	1.315042	0.188511	-0.01605	0.081495
Manhattan	-0.083101	0.025637	-3.241465	0.001191	-0.133352	-0.032851
Queens	-0.026604	0.025166	-1.057138	0.290461	-0.075933	0.022724
Bronx	0.004188	0.02649	0.158079	0.874396	-0.047735	0.05611
Inattention	0.05497	0.013534	4.061488	4.9E-05	0.028441	0.081499
FollowingTooClosely	0.077749	0.026336	2.952219	0.003159	0.026129	0.129369
FailureToYield	0.257621	0.020925	12.311502	0	0.216606	0.298636
UnspecifiedFactor	-0.034781	0.011559	-3.009085	0.002624	-0.057437	-0.012125
Sedan	0.011653	0.012247	0.951513	0.341356	-0.012352	0.035657
SUV	8.9E-05	0.01188	0.007483	0.99403	-0.023197	0.023375
Pickup	-0.060353	0.033921	-1.779203	0.075222	-0.126841	0.006136
Motorcycle	0.325619	0.051453	6.328464	0	0.224766	0.426471

Table 3. OLS estimates, standard errors, t-statistics, p-values, and 95% confidence intervals for all 15 model parameters.

9.

Fitted Model and Estimates on the Real Data

Using the full analysis sample, the estimated coefficient vector is: $\hat{\beta} = (0.3668,\ 0.0820,\ -0.1050,\ 0.0327,\ -0.0831,\ -0.0266,\ 0.0042,\ 0.0550,\ 0.0777,\ 0.2576,\ -0.0348,\ 0.0117,\ 0.0001,\ -0.0604,\ 0.3256)$ with $\hat{\sigma} = 0.6719, R^2 = 0.0238, and\ and\ adjusted\ R^2 = 0.0231$. Table 2 reports all estimates.

Parameter	Bronx	Brooklyn	FailureToYield	FollowingTooClosely	Inattention	Intercept	Manhattan
Estimate	0.004188	0.032722	0.257621	0.077749	0.05497	0.366793	-0.083101

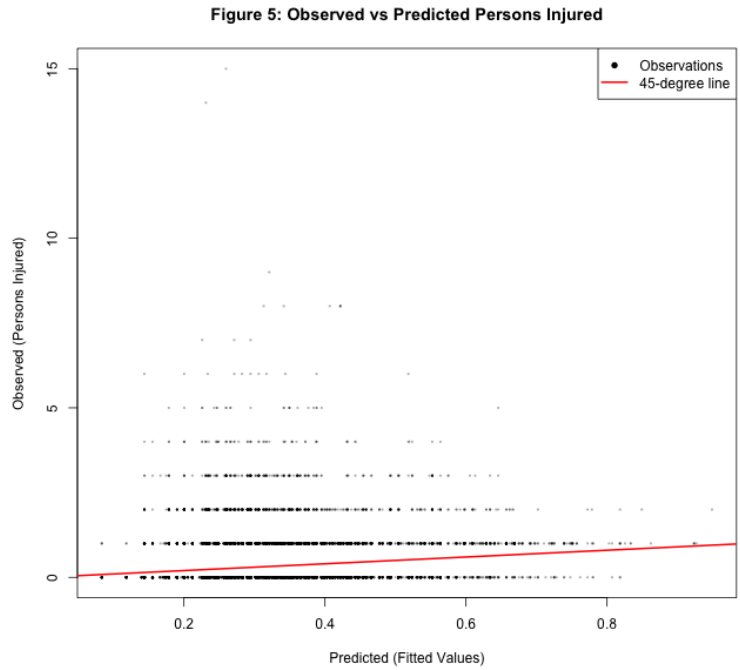
Motorcycle	MultiVehicle	Night	Pickup	Queens	Sedan	SUV	UnspecifiedFactor
0.325619	-0.105036	0.081951	-0.06035	-0.02660	0.011653	8.9E-05	-0.034781

Table 2. Estimated regression coefficients from OLS fit (n=19,999).

I believe several estimates are noteworthy here, the Night coefficient (+0.0820) means nitetime crashes are associated with about 0.082 additional injured persons on average, controlling for other variables. The FailureToYield coefficient (+0.2576) is also one of the larger positive effects, it reflects dangerous conflict at right-of-way violations. The Motorcycle coefficient (+0.3256) is the largest positive, consistent with a litany of reasons why my mom says I can't own a motorcycle.

The MultiVehicle coefficient (−0.1050) is negative after adjusting for the other predictors. This does not mean multi-vehicle crashes are safer; it reflects the conditional mean relationship once borough, time, contributing factor, and vehicle type are held fixed, showing adjusted estimates can differ from raw comparisons.

Figure 5 shows observed vs. predicted values. Most of the fitted values are compacted into ~ 0.2–0.8 while observations range 0–15. The $R^2 = 0.0238$ is small but not surprising. So many important determinants of injury severity (speed, impact angle, weather, impairment, seatbelt use, etc.) are unobserved/unrecorded. The model's value is in estimating interpretable directional relationships.

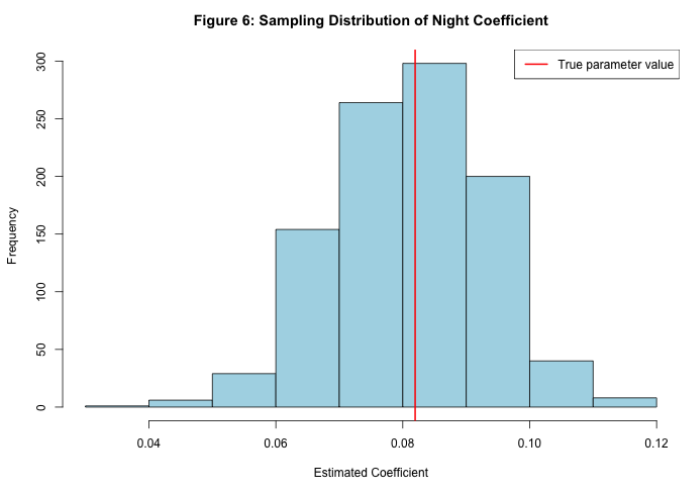


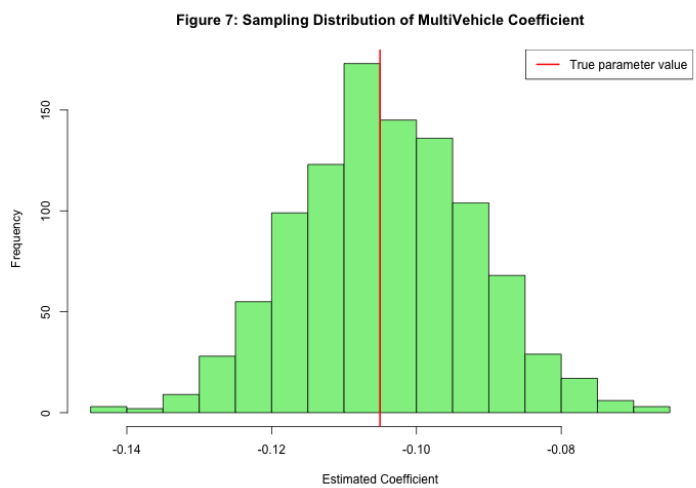
10. Broader Implications

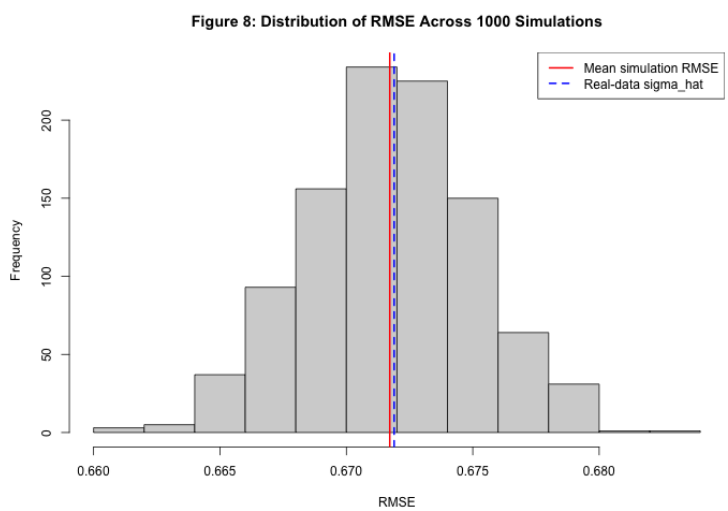
This fitted model suggests crash injury severity in New York City is analytically related to atleast these identifiable features. The positive effects for Night, FailureToYield, and Motorcycle support the view that reduced visibility, risky intersection behavior, and limited vehicle protection are each associated with greater average injury severity. This is reasonable and consistent with traffic-safety science.

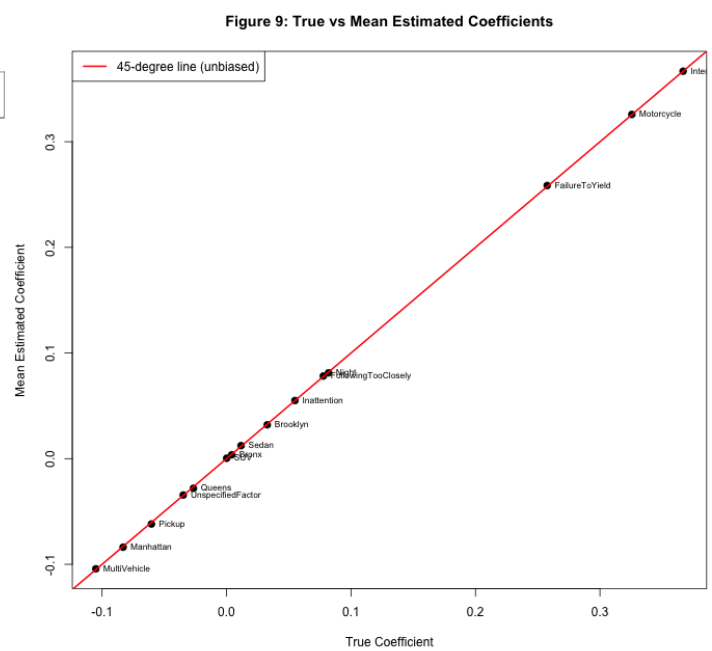
Nighttime driving may deserve special attention in identifying and planning from a public-safety perspective. The association for failure-to-yield crashes identifies intersection behavior as a target for driver education and or increased enforcement. The motorcycle finding highlights the idea that motorcycle crashes carry higher injury risk.

The model is observational and omits many relevant variables so these findings should not be interpreted as strong causal claims. Yet, it does provide a useful picture of the conditional means relationship in crash severity. It is consistent with established knowledge in the transportation safety research: Motorcycle involved accidents yields strong predictor of injury.









These simulation plots confirm: (1) that the sampling distributions for estimated coefficients are symmetric, and they are approximately normal (Figures 6–7); (2) distributions are all centered on the true parameter values; (3) RMSE is narrowly distributed and stable (Figure 8); (4) all 15 of the coefficients are recovered without systematic bias (Figure 9).

11. Limitations

1. The response is a non-negative integer with many event/entries at zero and a strong right-skewed distribution. A Poisson or negative binomial count model could be more appropriate. The linear model is used as an approximation to the conditional mean.

2. Speed, impact angle, weather, driver impairment, seatbelt use, road geometry, and many other factors are not listed. These more than likely correlate with included predictors, so estimated coefficients may reflect omitted-variable bias rather than the direct effects.

3. Contributing factors and vehicle types are reduced to a few indicators, losing accuracy. The R^2 =0.0238 confirms the model explains only a very small share of variation, yet, it remains valuable for inference as this data is being used successfully by New York City data scientist.

4. The simulation evaluates OLS only under the assumption that the fitted linear model is the true DGP. This verifies the algorithm, but it does not prove the linear model is the most realistic model for the actual crash process.

Data citation: NYC OpenData. *Motor Vehicle Collisions – Crashes*. New York City Department of Transportation. Available at:

https://data.cityofnewyork.us/Public-Safety/Motor-Vehicle-Collisions-Crashes/h9gi-nx95 (accessed March 2026).